idea TRIBE 2026

GOVERNMENT ENGINEERING COLLEGE PALAMU

IDEA PITCHING COMPETITION

FarmTrace

by Team Ekagra

Low-Cost IoT + Blockchain Nodes for Farm-to-Fork Traceability



PROBLEM

The problem we are targeting :-
Lack of real-time shipment condition monitoring and reliable farm-to-market traceability leads to post-harvest losses, quality disputes, and reduced trust in agricultural supply chains.





- **No real-time visibility:** Temperature, humidity, and transportation conditions may go unmonitored during transit.
- **Post-harvest losses:** Poor handling and unsuitable storage conditions can damage perishable produce.
- **Lack of traceability:** Difficult to track produce batches from collection to final delivery.
- **Quality disputes:** Buyers and suppliers often lack reliable shipment-condition records when damage occurs.

Unmonitered shipment conditions

Fragmented shipment records

Connectivity gaps

Cost of Uncertainty

6.02–15.05%
Estimated post-harvest losses in fruits

4.87–11.61%
Estimated post-harvest losses in vegetables

Source: NABCONS 2022 study, cited in the Ministry of Food Processing Industries Annual Report 2024–25.

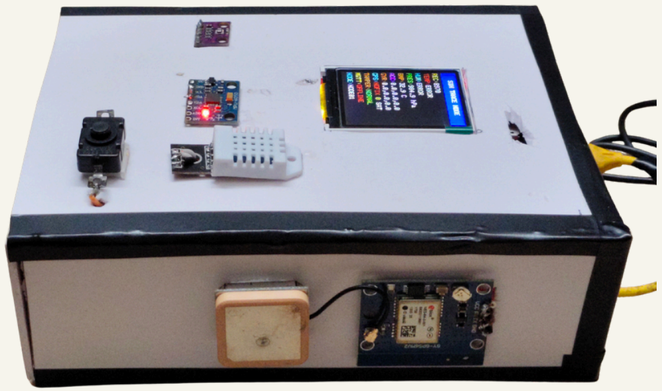
PROPOSED SOLUTION

FarmTrace provides a low-cost IoT and blockchain-based traceability system that monitors shipment conditions in real time and maintains tamper-evident records from farm to market.

- Real-time Monitoring
- GPS Tracking
- Tamper Evident Record
- QR based traceability
- Reduced post harvest losses
- Improve Transparency
- Low cost & reusable

FarmTrace Node





Our Prototype

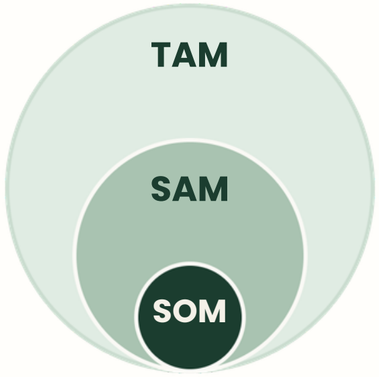


TARGET MARKET & CUSTOMERS

A ₹540 Cr market; we start with exporters and packhouses

Affordable 4G-enabled shipment monitoring and digital traceability for agricultural supply chains.

MARKET SIZE PER YEAR



₹540 Cr
TAM · total addressable / year

₹27 Cr
SAM · serviceable / year (5% of TAM)

₹1.62 Cr
SOM target · ARR by month 18 (6% of SAM)

Circles not to scale. Figures are preliminary estimates; a documented bottom-up sizing method is in progress.

WHO NEEDS FARMTRACE?

PRIMARY

Exporters & packhouses

Shipment-condition records, traceability reports and evidence for quality disputes.

NEXT

Cold-chain & logistics operators

Visibility into shipment conditions across transport routes.

NEXT

FPOs & agri aggregators

Affordable batch records and monitoring from collection to dispatch.

CUSTOMER NEEDS & LAUNCH STRATEGY

Core needs: shipment-condition monitoring, reliable digital records, QR lot traceability, easier quality-dispute investigation.

Initial geography: Jharkhand, Bihar and Eastern India.

Go-to-market: start with exporters and packhouses; expand through FPOs and logistics partners.

Who pays: the organization that gains shipment visibility and holds the budget, not necessarily the individual farmer.

REVENUE MODEL

REVENUE MODEL · PROPOSED PRICING TO VALIDATE

Two ways to pay: buy the node, or rent it with service

OPTION A · OWN THE HARDWARE

Buy & Subscribe

Customer owns the node; pays monthly for dashboard, shipment records, reports, alerts and software support.

₹8,499
one-time, per node

₹1,200 / mo
dashboard · ₹14,400/yr per node

OPTION B · RENT + SERVICE

FarmTrace Managed Rental

Node, dashboard, shipment history, reports, alerts and defined maintenance + support.

₹4,000 / mo
monthly plan, per node

₹36,000 / yr
annual, per node · 25% off

A · PER NODE, YEAR 1

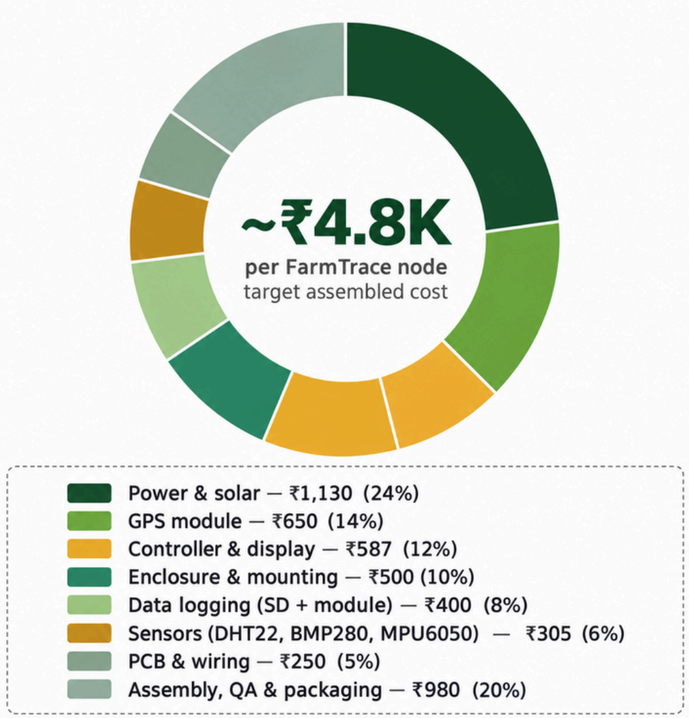
Node selling price	₹8,499
Dashboard ₹1,200 × 12	₹14,400
First-year revenue	₹22,899
Manufacturing + assembly	−₹4,800
Service delivery cost	−₹1,200
Gross contribution	₹16,899
Margin	73.8%

₹3,699 from hardware + ₹13,200 a year from the dashboard.

B · PER NODE, ANNUAL RENTAL

Annual rental revenue	₹36,000
Manufacturing, upfront	−₹4,800
Service, data, cloud	−₹1,200
Maintenance reserve	−₹1,000
Return logistics, support	−₹500
Cash contribution	₹28,500
Margin	79.2%

Monthly plan: ₹48,000 a year → ₹40,500 contribution (84.4%).



Cost targets to validate: ₹4,800 build, ₹1,200/yr service delivery (our cost, not the ₹1,200/mo price). Deposit ₹5,000/node, refundable, not revenue. Contribution, not net profit: excludes salaries, marketing, R&D, taxes, sales costs and repairs.

₹1,00,000 SEED GRANT · PROPOSED ALLOCATION

Five industrial 4G nodes, tested in the field

One standard node: every unit is 4G-enabled with industrial-grade sensors. No separate Wi-Fi-only version.

5
Industrial 4G nodes
One standardized hardware version

2–3
Pilot partners
FPOs, packhouses or exporters

10–20
Shipment trials
Real-world monitoring tests

1
Commercial cost model
Verified BOM and operating costs

SPLIT OF ₹1,00,000



Hardware	₹70,000 · 70%
Software, validation, pilot	₹30,000 · 30%

DETAILED BUDGET

1. 5 industrial 4G nodes · 5 × ₹14,000	₹70,000
2. Sensor accuracy validation	₹6,000
3. Tamper protection and mounting	₹4,000
4. Battery and solar testing	₹4,000
5. Dashboard development	₹6,000
6. SIM, data and cloud testing	₹3,000
7. Field pilot deployment, 2–3 organizations	₹5,000
8. Spare parts and contingency	₹2,000
Total	₹1,00,000

₹14,000 per node is a target budget, not a supplier quotation; if quotes come in higher, the node count or another line will be adjusted. Proposed allocation, not completed results.

COMPETITIVE LANDSCAPE

Brand	Focus Area	Difference from FarmTrace	Limitations
SENSITECH Cold Chain Visibility	Temperature monitoring and cold-chain visibility	→ FarmTrace combines affordable IoT hardware with blockchain-backed records and QR traceability.	• High hardware and subscription costs. • Limited focus on agricultural supply chains. • No blockchain-based records or QR traceability.
tive Real-Time Visibility	Real-time shipment location and condition tracking	→ FarmTrace focuses on a low-cost, reusable solution for agricultural supply chains.	• Higher cost per device. • Requires continuous internet connectivity. • Limited blockchain integration and traceability features.
Controlant Supply Chain Intelligence	Cold-chain monitoring and analytics	→ FarmTrace aims to offer simpler, more affordable monitoring for smaller regional businesses.	• High setup and operational costs. • Complex platform and integration. • Focused mainly on large enterprises and global supply chains.
Traditional Data Loggers	Recording temperature and humidity	→ FarmTrace combines sensor records with shipment traceability and blockchain-supported data integrity.	• No real-time location tracking. • No tamper-evident records. • Limited to individual data points (no end-to-end traceability).
Manual / Paper Tracking	Recording shipment details manually	→ FarmTrace provides digital records, QR-based access, and improved auditability.	• Prone to human error and fraud. • No real-time updates. • No integration with analytics or blockchain.
FarmTrace Trace Today, Trust Tomorrow	Farm-to-market monitoring and traceability	→ Low-cost IoT, offline data storage, reusable nodes, QR access, and blockchain-backed tamper evidence.	• Currently focused on regional markets (Jharkhand, Bihar, Eastern India). • Limited scale and brand recognition (early stage). • Requires user adoption and network growth for full potential.

TECHNICAL APPROACH

* **Low-Cost IoT:** Affordable ESP8266 and sensors monitor shipment conditions.

* **Blockchain Security:** SHA-256 hash chaining and blockchain anchoring help detect record tampering.

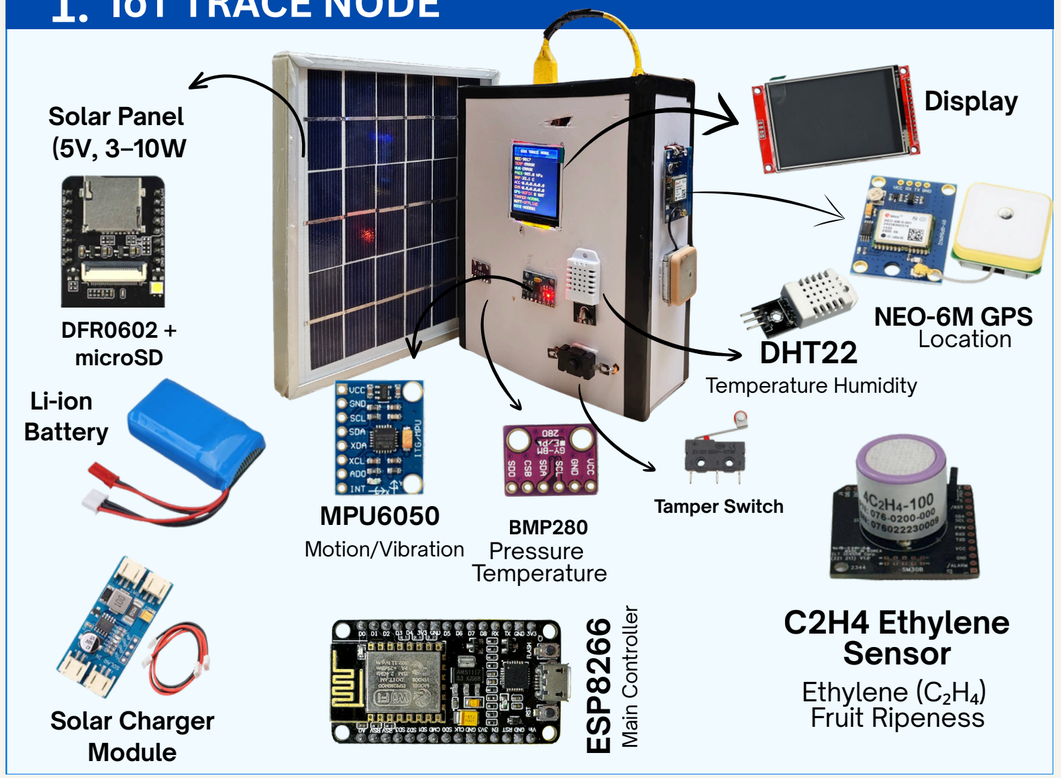
* **Offline Storage:** Saves data during network interruptions.

* **QR Traceability:** Buyers can access shipment history through QR codes.

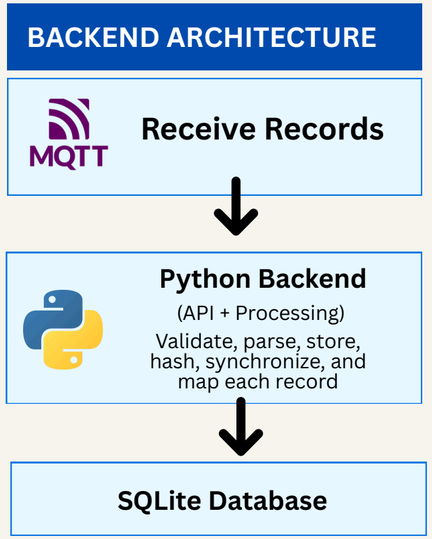
* **Reusable Nodes:** One device can monitor multiple shipments, reducing costs.

* **Modular Design:** Easy maintenance and affordable upgrades.

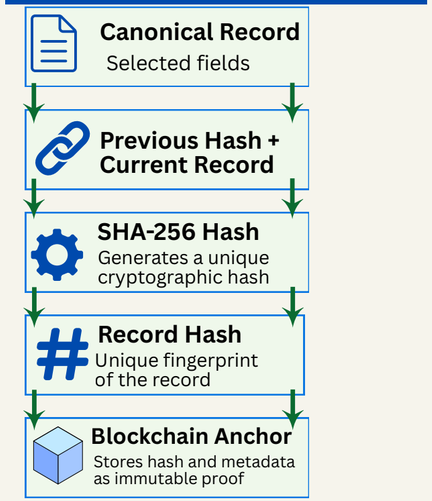
1. IoT TRACE NODE



BACKEND ARCHITECTURE



BLOCKCHAIN & DATA INTEGRITY



TRACEABILITY & TRANSPARENCY

Monitoring Dashboard
Authorized Stakeholders

QR Code Access
View product journey and verification status

Multi-Participant Access
Farm, Processor, Transporter, Retailer

Audit & Compliance
Regulators / Certifiers

Team Ekagra



Vimal Kumar Singh

Team Leader



Swapnil Pradhan



Supriyo Choudhary



Piyush Pushkar Pandey



Ujjawal